

Test Specification

Supplier Reference | Rev. 00 - June 2026 |

Test Sequence

Visual Inspection

Verify HMI display is not damaged.

Applied Voltage Test (Hipot)

Short-circuit all power terminals: input, DC bus, output.

Apply DC voltage between shorted terminals and chassis ground:

220 V models (T2): 2100 Vdc / 2 s

380-480 V models (T4): 2800 Vdc / 2 s

500-600 V models (T5): 3111 Vdc / 2 s

Pass: no insulation breakdown, leakage current max 4 mA.

Voltage must ramp up and down over 1 s at start and end.

Control Power Tests

External 24 V supply: power inverter via 24 V +/-10% -- HMI must start without faults.

Internal 24 V output: apply 250 mA load -- verify 21.6 to 26.4 V.

Firmware and Model Programming

Load firmware and model configuration via Ethernet or RS-485.

Power-cycle the inverter -- must restart without errors.

Confirm model identifier matches the unit under test.

Serial Number and MAC Address

Write serial number and Ethernet MAC address via Modbus during the test routine.

Communication Interfaces

Ethernet Port A: Modbus TCP read must succeed.

Ethernet Port B: same test repeated.

RS-485: Modbus RTU -- Read-Device-ID must return correct identification.

Analog Inputs

Current mode 0 to 20 mA: verify reading at 0 / 4 / 10 / 16 / 20 mA -- tolerance +/-2%.

Voltage mode 0 to 10 V: verify reading at 0 / 2 / 5 / 8 / 10 V -- tolerance +/-2%.

Analog Outputs

Voltage mode 0 to 10 V: verify output at 0 / 20 / 50 / 80 / 100% -- tolerance +/-0.2 V.

Current mode 0 to 20 mA: verify output at 0 / 20 / 50 / 80 / 100% -- tolerance +/-0.4 mA.

Digital Inputs (DI1 to DI6)

Test all 6 inputs in source (PNP) and sink (NPN) configurations.

Pass: parameter status correctly reflects each input state.

Digital Output (DO1)

OFF state: load voltage must be 0 V +/-0.2 V.

ON state: load voltage must be 21.6 to 26.4 V.

Control Supply Voltages

24 V IO supply: 21.6 to 26.4 V (20.0 to 26.4 V when using external supply).

RTC battery: 2.85 to 3.15 V.

RTC Test

Set current date and time via parameter.

Verify after 5 s and after a power cycle -- tolerance +/-1 s.

Fan Speed

Enable fans via parameter and verify speed for each mechanical size -- tolerance +/-10%.

Motor Operation

Connect a motor or a electronic load of the same rated power as the inverter under test.

Run across full speed range: 3 to 60 Hz.

Pass: current reading within +/-5%.

STO- Safe Torque Off

OSSD mode: with STO signals OFF, inverter must enter STO state without faults.

Turn ON STO signals -- inverter goes to Ready, motor runs normally.

Trip STO while motor is running -- motor must stop with no fault generated.

Dry contact mode: repeat the same sequence using relay contacts driven by SP1/SP2 test pulses.

Dynamic Braking (DB models only)

Trigger braking circuit -- verify if there is voltage under the braking connectors.

Protection Tests

Ground fault: short phase W to the ground with a contactor just a pulse – fault must be indicated.

Short circuit: short phases U and V with a contactor just a pulse-- fault must be indicated.

After each test: remove fault, release inverter, verify normal operation resumes.

Load Test

Run at 55 Hz, ambient temperature 50 °C, apply current derating per user manual.

Pass: continuous operation for at least 20 minutes without any fault.

End of Test

Load factory defaults.

De-energize the inverter.